MATH 6660 — Homework 4

Fall 2026

Problem 1

Problem. Download the PyDistMesh mesh generator. You can find all the necessary links from the Math 6660 Canvas page.

- Downloaded from Canvas. PyDistMesh is B. Froehle's Python port (2012) of P.-O. Persson's DistMesh (2004). Nothing in it was edited.

file	role
`distmesh2d.py`	`p,t = distmesh2d(fd,fh,h0,bbox,pfix)`: level set `fd`, relative size `fh`, edge length `h0`; returns nodes `p` ($N \times 2$) and triangles `t` ($M \times 3$)
`distance_functions.py`	level sets (`dcircle`, `drectangle`, ...) and Booleans: `dunion` = min, `dintersect` = max, `ddiff`$(d_1, d_2) = \max(d_1, -d_2)$
`boundary.py`	`bnd2d`: re-triangulates, marks nodes with $\lvert\phi\rvert \le 10^{-8}$ as boundary and puts them last, returns the areas
`assemble.py`	the cG(1) code used in Problem 3
`example1.py`, `example2.py`, `poisson_example.py`	unit disc; square with a hole, graded; $-\Delta u = 1$ on the latter

Table 1: The PyDistMesh bundle as downloaded (plus two helper files, `utils.py` and `mlcompat.py`).

Step 1 — what in the code is the lecture [Lecture 13, Part IV, eqs. (1.278)–(1.282)]

- `fd` is the lecture's level set ϕ: $\partial\Omega = \{\phi = 0\}$, $\Omega = \{\phi < 0\}$. The Booleans are the lecture's min (union) and max (intersection); a set difference is $\max(\phi_1, -\phi_2)$.
- Points become triangles by the lecture's two Python lines: a Delaunay triangulation from `scipy.spatial`, then only the triangles whose centroid has $\phi < 0$ are kept.
- Areas are the lecture's $\frac{1}{2}|\underline{v}_1 \times \underline{v}_2|$. Boundary nodes go last, so $u = 0$ is imposed by assembling only the first `num_int_nodes` rows and columns.

NOT FROM LECTURE — where the nodes come from

The lecture triangulates a given point set. DistMesh generates the points: a lattice of spacing h_0 cut to $\phi < 0$, then relaxed by treating every Delaunay edge as a compressed bar of rest length $1.2\,$`fh`, re-triangulating as nodes move and projecting escaped nodes back onto $\phi = 0$. Persson and Strang, *SIAM Review* 46 (2004).

Step 2 — it runs

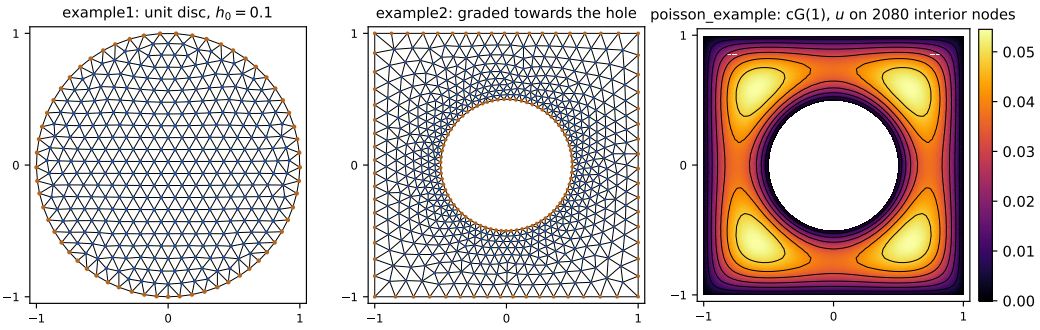


Figure 1: The three bundled examples, run unchanged. Interior nodes blue, boundary nodes brown. Right: the cG(1) solution of $-\Delta u = 1$, $u = 0$ on $\partial\Omega$.

example	h_0	interior	boundary	elements	mesh area	exact	$\max_j h_j/\rho_j$
unit disc, uniform	0.10	301	61	661	3.1359	3.1416	2.52
square with hole, graded	0.04	532	142	1206	3.2156	3.2146	2.83
square with hole, uniform, Poisson	0.04	2080	291	4451	3.2155	3.2146	2.64

Table 2: The three runs. Exact area: π for the disc, $4 - \pi/4$ for the square with a hole of radius $1/2$. Last column: the lecture's regularity ratio (1.260), longest edge over inscribed diameter; an equilateral triangle gives $\sqrt{3} = 1.73$.

- Areas agree to 0.2%; the disc comes out smaller and the holed square larger, since chords cut a sliver off a convex boundary and add one at a concave one. $\max_j h_j/\rho_j \le 2.83$: no slivers.

Problem 2

Problem. Use this code to generate a mesh for a star-shaped region whose boundary is the zero contour of $\phi(r,\theta) = r - 0.75 - 0.25\sin(5\theta)$.

Step 1 — the level set in x, y [Lecture 13, eqs. (1.279)–(1.280)]

- `fd` receives Cartesian points, so with $r = \sqrt{x^2+y^2}$, $\theta = \operatorname{atan2}(y,x)$:

$$\begin{gathered}\phi(x,y) = \sqrt{x^2+y^2} - 0.75 - 0.25\sin\big(5\,\operatorname{atan2}(y,x)\big),\\ \phi < 0 \iff r < R(\theta) := 0.75 + 0.25\sin 5\theta.\end{gathered} \tag{2.1}$$

  $0.5 \le R \le 1$: five tips at $r = 1$, five valleys at $r = 0.5$; the box $(-1,1)^2$ contains Ω.

```
fd = lambda p: np.sqrt(p[:,0]**2 + p[:,1]**2) - 0.75 \
               - 0.25*np.sin(5*np.arctan2(p[:,1], p[:,0]))
p, t = dm.distmesh2d(fd, distfunc.huniform, h0, (-1,-1,1,1))
p, t, nin, nbd, area = bnd.bnd2d(fd, p, t, -1.0, -1.0, 1.0, 1.0)
```

- Exact area, to check the mesh against:

$$|\Omega| = \int_0^{2\pi}\!\!\int_0^{R(\theta)} r\,\mathrm{d}r\,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}\Big[0.75^2 + 1.5\cdot 0.25\underbrace{\sin 5\theta}_{\int=0} + 0.25^2\underbrace{\sin^2 5\theta}_{\int=\pi}\Big]\mathrm{d}\theta = \frac{19\pi}{32} = 1.86532. \tag{2.2}$$

Step 2 — the boundary nodes, and a fix

- `bnd2d` calls a node *boundary* only if $|\phi| \le 10^{-8}$, and Problem 3's solver sets $u = 0$ only there. So every node on a boundary edge (an edge of one triangle only) must pass that test.
- **It does not.** At $h_0 = 0.05$ only 72 of the 138 nodes on boundary edges pass (Table 3, last column). The 66 others sit just inside, $-1.0 \times 10^{-4} \le \phi < -10^{-8}$, all on the inner part of the boundary, $r \le 0.78$, around the five valleys (Figure 2, red).
- **Why.** `distmesh2d` moves a node that left Ω back by $\boldsymbol{p} \leftarrow \boldsymbol{p} - \phi\nabla\phi/|\nabla\phi|^2$, repeated *while* $\phi > 10^{-14}$. One step lands exactly on $\phi = 0$ only if ϕ is linear along $\nabla\phi$, as for $\phi = r - 1$ in the bundle's examples. Here

$$\nabla\phi = \underbrace{\boldsymbol{e}_r}_{\partial\phi/\partial r} - \underbrace{\frac{1.25\cos 5\theta}{r}\,\boldsymbol{e}_\theta}_{\frac{1}{r}\partial\phi/\partial\theta} \tag{2.3}$$

  is not radial, so a step can land inside, and then the loop stops with $\phi < 0$.
- **Fix** (the bundle is not edited). Before `bnd2d`, move every node on a boundary edge radially onto the curve:

$$(x, y) \;\leftarrow\; R(\theta)\,(\cos\theta, \sin\theta), \qquad \text{so} \quad \phi = R(\theta) - R(\theta) = 0. \tag{2.4}$$

  The nodes move at most $10^{-4} \ll h_0$. Afterwards every boundary node passes (Table 3).

Step 3 — the mesh [Lecture 13, eq. (1.260): regular triangulation]

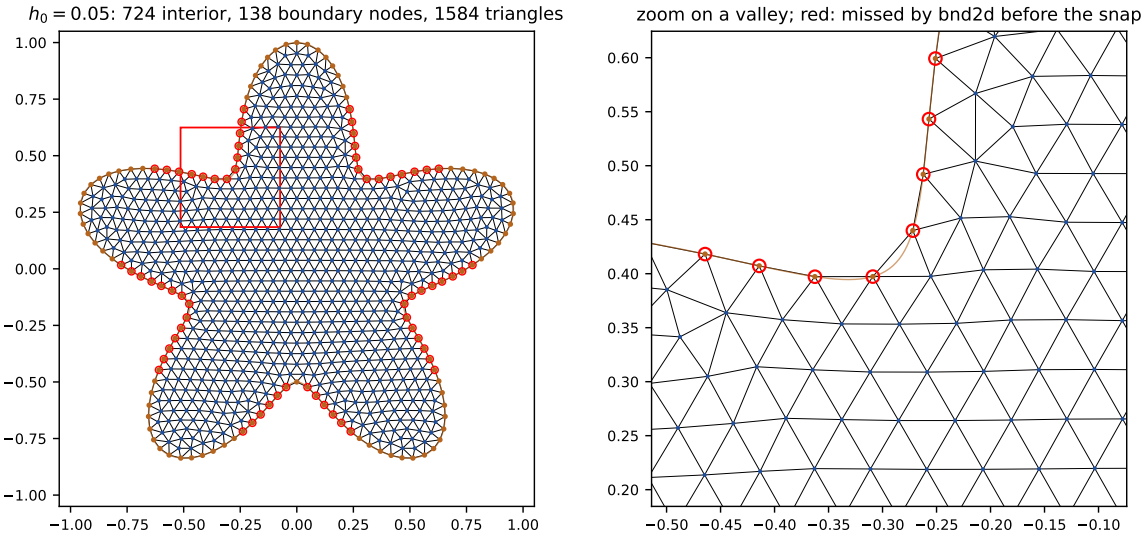


Figure 2: The star mesh at $h_0 = 0.05$, after the fix (2.4). Blue: interior nodes; brown: boundary nodes; the curve is $\phi = 0$. Red circles: the boundary nodes that `bnd2d` classed as interior before the fix.

h_0	interior	boundary	elements	mesh area	rel. error	$\max_j h_j/\rho_j$	flagged: raw → snapped
0.100	149	66	362	1.86001	-2.8×10^{-3}	2.57	32/66 → 66/66
0.050	724	138	1584	1.86432	-5.4×10^{-4}	2.46	72/138 → 138/138
0.025	3167	282	6614	1.86498	-1.9×10^{-4}	2.45	140/282 → 282/282

Table 3: Star meshes. Exact area $19\pi/32 = 1.86532$ (2.2). Last column: nodes `bnd2d` marks as boundary, out of the nodes on boundary edges, before and after (2.4).

- The area error falls from 0.3% to 0.02% as h_0 goes from 0.1 to 0.025.
- $\max_j h_j/\rho_j \le 2.57$ on every mesh (equilateral: 1.73): regular in the sense of (1.260), no slivers in the valleys.

Problem 3

Problem. Use the CG(1) method developed in class and provided on the Math 6660 Canvas page to solve $-u_{,xx} - u_{,yy} = 1$ in Ω, $u = 0$ on $\partial\Omega$, on the star of Problem 2.

Step 1 — the method [Lecture 13, eqs. (1.236) and (1.253)–(1.255)]

- Weak form: find $u \in H_0^1(\Omega)$ with $\iint_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y$ for all v, $f = 1$.
- V_h: continuous, linear on each triangle of Problem 2, zero at the boundary nodes. Basis: the hat φ_i of each interior node. $U = \sum_i \xi_i \varphi_i$ and $\boldsymbol{A}\boldsymbol{\xi} = \boldsymbol{b}$ with $A_{ij} = \iint \nabla\varphi_i \cdot \nabla\varphi_j$, $b_j = \iint f\varphi_j$.

Step 2 — what `assemble.py` computes, triangle by triangle

- On a triangle K with vertices (x_l, y_l), $l = 1, 2, 3$, the hat of vertex k is $a_k + b_k x + c_k y$ with value δ_{kl} at vertex l:

$$\underbrace{\begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}}_{\boldsymbol{M}} \begin{bmatrix} a_k \\ b_k \\ c_k \end{bmatrix} = \boldsymbol{e}_k \quad \Longrightarrow \quad (a_k, b_k, c_k)^T = \text{column } k \text{ of } \boldsymbol{M}^{-1}, \tag{3.1}$$

  and $\nabla\varphi_k = (b_k, c_k)$ on K.
- The gradients are constant on K, so the stiffness integral is exact:

$$\iint_K \nabla\varphi_j \cdot \nabla\varphi_k \,\mathrm{d}x\,\mathrm{d}y = |K|\,(b_j b_k + c_j c_k). \tag{3.2}$$

- The load uses the vertex rule, Lecture 4's trapezoidal rule on a triangle:

$$\iint_K f\varphi_k \,\mathrm{d}x\,\mathrm{d}y \approx \frac{|K|}{3} \sum_{l=1}^{3} f(x_l, y_l)\, \underbrace{\varphi_k(x_l, y_l)}_{\delta_{kl}} = \frac{|K|}{3}\, f(x_k, y_k). \tag{3.3}$$

  For $f \equiv 1$ it is exact. φ_k is linear, so its mean over K is its value at the centroid, the average of the vertex values, $(1 + 0 + 0)/3$.
- Only nodes with index below `num_int_nodes` get a row and a column: $u = 0$ is built into V_h. The assembled $\boldsymbol{A}$ is symmetric to round-off ($\max|\boldsymbol{A} - \boldsymbol{A}^T| = 0$).

Step 3 — check on a domain with a known answer

- Unit disc: $u = \frac{1}{4}(1 - x^2 - y^2)$ has $u_{,xx} = u_{,yy} = -\frac{1}{2}$, so $-\Delta u = 1$, and $u = 0$ at $r = 1$. Same pipeline, no fix needed: for $\phi = r - 1$ every boundary node passes.

h_0	interior nodes	$\max U$	$\max_i \lvert U_i - u(x_i, y_i)\rvert$	rate
0.100	301	0.24983	4.03×10^{-4}	—
0.050	1324	0.24984	9.63×10^{-5}	2.06
0.025	5559	0.25000	2.37×10^{-5}	2.02

Table 4: The Canvas cG(1) code on the unit disc, $-\Delta u = 1$: largest nodal error against $\frac{1}{4}(1 - x^2 - y^2)$. It falls as h_0^2.

Step 4 — the star

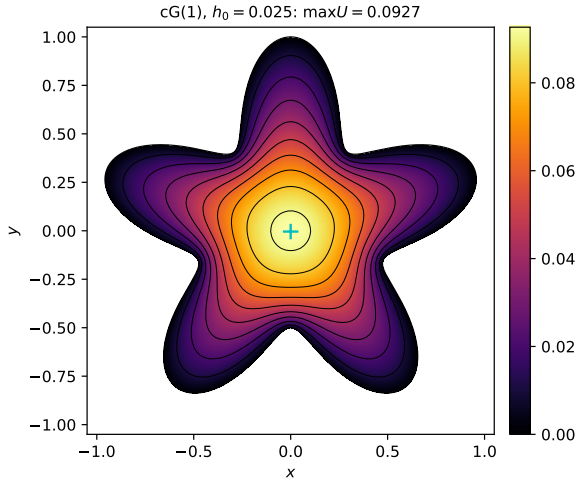


Figure 3: cG(1) solution of $-\Delta u = 1$, $u = 0$ on the star, $h_0 = 0.025$ (3167 unknowns), with ten contours. Cross: the largest nodal value.

h_0	unknowns	$\max U$	at (x, y)	$\iint_{\Omega_h} U\,dx\,dy$	$\max U$, raw mesh
0.100	149	0.09233	(0.000, 0.032)	0.06619	0.23841
0.050	724	0.09256	(−0.026, −0.008)	0.06747	0.22573
0.025	3167	0.09266	(0.000, −0.004)	0.06764	0.22951

Table 5: The star under refinement. "Raw mesh": the same nodes without the fix (2.4) of Problem 2.

- $\max U = 0.0927$, at the node nearest the centre — where the five-fold symmetry puts it — and stable to three digits from $h_0 = 0.05$.
- Without the fix, $\max U \approx 0.23$: the nodes missed in Problem 2 are left free, so along the valleys the code imposes no condition instead of $u = 0$. The solution is then wrong.

Result. On the star, $\max u \approx 0.0927$ at the centre and $\iint_\Omega u\,\mathrm{d}x\,\mathrm{d}y \approx 0.0676$ (Table 5, Figure 3).

Mechanics reading. A membrane of unit tension under unit pressure, rim clamped. The disc inscribed in the star ($r = 0.5$) deflects $1/16 = 0.0625$ at its centre, the circumscribed disc ($r = 1$) $1/4$; the star sits between, at 0.0927. The raw-mesh 0.23 is almost the circumscribed value: a membrane released along its valleys.

Relation to the lectures. Lecture 13, Part III, does exactly this problem with $u = 0$ on $\partial\Omega$: product rule, divergence theorem, $V = H_0^1$, and the four Lax–Milgram checks with a Poincaré constant left unnamed. Two things change here. The boundary value is $g \neq 0$, which is Lecture 5's nonhomogeneous Dirichlet case (trial set $\neq$ test space, lifting). And the elements are squares with the bilinear interpolant the problem gives; their basis turns out to be products of Lecture 4's 1D hats. The Poincaré constant for the square is computed by Lecture 8's 1D argument.

Problem 4(a)

Problem. Recast this problem as a variational problem. Clearly state the test and trial function spaces.

The strong problem is

$$-\nabla \cdot \nabla u = -u_{,xx} - u_{,yy} = f(x,y) \quad \text{in } \Omega = (-1,1) \times (-1,1), \qquad u = g \quad \text{on } \partial\Omega. \tag{4a.1}$$

Step 1 — multiply by v, integrate, move one derivative [Lecture 13, Part III: product rule, then divergence theorem]

- Multiply (4a.1) by a test function v and integrate over Ω:

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.2}$$

- Product rule, $\nabla \cdot (v\nabla u) = \nabla v \cdot \nabla u + v\,\Delta u$, so the left side of (4a.2) is

$$-\iint_\Omega v\,\Delta u\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla \cdot (v\nabla u)\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.3}$$

- Divergence theorem on the last term, $\boldsymbol{n}$ the outward unit normal: $\iint_\Omega \nabla\cdot(v\nabla u)\,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial\Omega} v\,\nabla u\cdot \boldsymbol{n}\,\mathrm{d}s$. Put this into (4a.3) and (4a.3) into (4a.2):

$$\iint_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x\,\mathrm{d}y \;-\; \oint_{\partial\Omega} v \underbrace{\nabla u \cdot \boldsymbol{n}}_{\text{normal flux, not given}} \mathrm{d}s \;=\; \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y. \tag{4a.4}$$

Step 2 — kill the boundary term [Lecture 5: Dirichlet = essential, the test function vanishes there]

- On all four sides the data is the *value* $u = g$. The flux $\nabla u \cdot \boldsymbol{n}$ there is not known — it is the reaction to holding u at g. So the boundary integral in (4a.4) cannot be evaluated.
- The only way to remove it is to demand $v = 0$ on $\partial\Omega$. Then (4a.4) becomes

$$\boxed{\iint_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y} \tag{4a.5}$$

Step 3 — the trial set and the test space [Lecture 5: $u(0) = \alpha \neq 0$ gives trial $V \neq$ test V_0; Lecture 13, eq. (1.254), for the 2D integrals]

- (4a.5) needs $\iint_\Omega |\nabla u|^2$ and $\iint_\Omega |\nabla v|^2$ finite, so both spaces are built on $\iint_\Omega \left[v^2 + |\nabla v|^2\right] \mathrm{d}x\,\mathrm{d}y < \infty$, as in Lecture 13.

- The unknown u must also satisfy the essential condition $u = g$; the test function must satisfy $v = 0$ (Step 2):

$$\boxed{\begin{aligned} &\text{trial:}\quad V_g = \Big\{ v : \iint_\Omega \big[v^2 + |\nabla v|^2\big]\,\mathrm{d}x\,\mathrm{d}y < \infty,\quad v = g \text{ on } \partial\Omega \Big\}, \\ &\text{test:}\quad V_0 = \Big\{ v : \iint_\Omega \big[v^2 + |\nabla v|^2\big]\,\mathrm{d}x\,\mathrm{d}y < \infty,\quad v = 0 \text{ on } \partial\Omega \Big\} = H_0^1(\Omega), \end{aligned}} \tag{4a.6}$$

  with $|\nabla v|^2 = (v_{,x})^2 + (v_{,y})^2$.
- They differ because $g \neq 0$ (in part (d), $g = 1$). Exactly Lecture 5's warning: V_g is not a vector space — $u_1 + u_2$ has boundary value $2g$ — while V_0 is.

(Variational problem) Find $u \in V_g$ such that

$$\iint_\Omega \nabla u \cdot \nabla v\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f v\,\mathrm{d}x\,\mathrm{d}y \qquad \text{for all } v \in V_0,$$

with trial set V_g and test space $V_0 = H_0^1(\Omega)$ from (4a.6).

Mechanics reading. A membrane with unit tension under pressure f, its rim held at height g. V_g: deflected shapes that honour the rim. V_0: virtual deflections, which may not move the rim. (4a.5) is virtual work; the rim reaction does no virtual work because $v = 0$ there.

Problem 4(b)

Problem. Show that the variational problem has a unique solution by showing that it meets all of the criteria of the Lax–Milgram theorem.

Step 1 — put the problem in Lax–Milgram's form [Lecture 13: u and v in the same Hilbert space; Lecture 5: lift the boundary data]

- Lax–Milgram (Lecture 13) asks for *one* Hilbert space V with $u, v \in V$. In 4(a), $u \in V_g$ but $v \in V_0$. Lecture 5's fix: split off a known function that carries the boundary data.
- Pick any $G \in V_g$. (If V_g were empty there would be no candidate u at all.) For part (d), $g = 1$ and $G \equiv 1$ works: $\iint_\Omega [1^2 + 0]\,\mathrm{d}x\,\mathrm{d}y = 4 < \infty$.
- Write $u = w + G$. Then $w = g - g = 0$ on $\partial\Omega$, so $w \in V_0$. Substitute in (4a.5) and move the known part to the right:

$$\underbrace{\iint_\Omega \nabla w \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{B(w,v)} = \underbrace{\iint_\Omega f v \,\mathrm{d}x\,\mathrm{d}y - \iint_\Omega \nabla G \cdot \nabla v \,\mathrm{d}x\,\mathrm{d}y}_{\tilde{L}(v)} \qquad \forall\, v \in V := V_0 = H_0^1(\Omega), \quad \text{(4b.1)}$$

  with the norm of Lecture 13, $\|v\|_V^2 = \|v\|^2 + \|\nabla v\|^2$, where $\|\cdot\|$ is the $L^2(\Omega)$ norm.

Step 2 — (S) and (CB) [Lecture 13, Part III, checks 1–2; Lecture 6: Cauchy]

- **(S)** $\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$ pointwise, so $B(w, v) = B(v, w)$.
- **(CB)**

$$|B(w,v)| \le \underbrace{\iint_\Omega |\nabla w|\,|\nabla v| \,\mathrm{d}x\,\mathrm{d}y}_{|\boldsymbol{a}\cdot\boldsymbol{b}| \le |\boldsymbol{a}||\boldsymbol{b}|} \le \underbrace{\|\nabla w\|\,\|\nabla v\|}_{\text{Cauchy}} \le \underbrace{\|w\|_V \|v\|_V}_{\|\nabla v\|^2 \le \|v\|^2 + \|\nabla v\|^2}, \qquad \beta^* = 1. \quad \text{(4b.2)}$$

Step 3 — Poincaré on the square [Lecture 8, eqs. (1.127)–(1.128): the same three lines, in x, for each fixed y]

- Take $v \in V_0$ and fix y. $v(-1, y) = 0$ on the left side, so

$$v(x,y) = \int_{-1}^{x} v_{,x}(s,y)\,\mathrm{d}s. \quad \text{(4b.3)}$$

- Bound it by the whole line and apply Cauchy to $1 \cdot |v_{,x}|$ on an interval of length 2:

$$|v(x,y)| \le \int_{-1}^{1} |v_{,x}(s,y)|\,\mathrm{d}s \le \underbrace{\Big(\int_{-1}^{1} 1^2\,\mathrm{d}s\Big)^{1/2}}_{\sqrt{2}} \Big(\int_{-1}^{1} v_{,x}(s,y)^2\,\mathrm{d}s\Big)^{1/2}. \quad \text{(4b.4)}$$

- Square (4b.4). The right side no longer depends on x, so integrating over $x \in (-1, 1)$ multiplies it by 2; then integrate over $y \in (-1, 1)$:

$$\boxed{\|v\|^2 = \iint_\Omega v^2 \,\mathrm{d}x\,\mathrm{d}y \le \underbrace{2 \cdot 2}_{x\text{-length} \times \text{(4b.4)}} \iint_\Omega (v_{,x})^2 \,\mathrm{d}x\,\mathrm{d}y \le 4\,\|\nabla v\|^2} \quad \text{(4b.5)}$$

  This is Lecture 13's multi-dimensional Poincaré, $\|v\| \le C\|\nabla v\|$, with $C = 2$.

Step 4 — (VE) and the bound on $\widetilde{L}$ [Lecture 13, Part III, checks 3–4]

- **(VE)** Split $B(v,v)$ in half and use (4b.5) on one half:

$$B(v,v) = \|\nabla v\|^2 = \tfrac{1}{2}\|\nabla v\|^2 + \underbrace{\tfrac{1}{2}\|\nabla v\|^2}_{\geq \frac{1}{8}\|v\|^2 \text{ by (4b.5)}} \geq \tfrac{1}{8}\big(\|\nabla v\|^2 + \|v\|^2\big) = \tfrac{1}{8}\|v\|_V^2, \qquad \beta_* = \tfrac{1}{8}. \tag{4b.6}$$

  Same as Lecture 13's $\beta_* = \frac{1}{2}\min(1/C^2, 1)$ with $C = 2$.

- $\widetilde{L}$ **bounded**, by Cauchy on each term of (4b.1):

$$|\widetilde{L}(v)| \leq \|f\|\,\|v\| + \|\nabla G\|\,\|\nabla v\| \leq \big(\|f\| + \|\nabla G\|\big)\|v\|_V. \tag{4b.7}$$

  This needs $\iint_\Omega f^2 < \infty$. In part (d), f is continuous on the closed square, $e^{\sin \pi x \sin \pi y} \leq e$ and the brace is at most $2+2+1+1 = 6$, so $|f| \leq 6\pi^2 e$; and $\nabla G = 0$.

condition	holds because	constant	equation
(S) symmetric	$\nabla w \cdot \nabla v = \nabla v \cdot \nabla w$	—	Step 2
(CB) bounded	Cauchy	$\beta^* = 1$	(4b.2)
(VE) V-elliptic	Poincaré, $C = 2$	$\beta_* = 1/8$	(4b.6)
$\widetilde{L}$ bounded	Cauchy	$\|f\| + \|\nabla G\|$	(4b.7)

Table 6: The four Lax–Milgram conditions for (4b.1) on $V = H_0^1\big((-1,1)^2\big)$.

Step 5 — conclusion, and uniqueness of u itself

- Table 6 meets every hypothesis of Lax–Milgram, so (4b.1) has a unique $w \in V_0$, with $\|w\|_V \leq \widetilde{L}/\beta_* = 8\big(\|f\| + \|\nabla G\|\big)$. Then $u = w + G \in V_g$ solves the variational problem of 4(a).
- u does not depend on which G was picked. If $u_1, u_2 \in V_g$ both solve it, $d = u_1 - u_2$ has boundary value $g - g = 0$, so $d \in V_0$, and subtracting the two problems gives $B(d,v) = 0$ for all $v \in V_0$. Take $v = d$ and use (4b.6):

$$\tfrac{1}{8}\|d\|_V^2 \leq B(d,d) = 0 \qquad \Longrightarrow \qquad d = 0, \quad u_1 = u_2. \tag{4b.8}$$

Conclusion. With $u = w + G$, the problem for w on $H_0^1(\Omega)$ satisfies (S), (CB) with $\beta^* = 1$, (VE) with $\beta_* = 1/8$, and $\widetilde{L}$ is bounded. By Lax–Milgram the variational problem of 4(a) has exactly one solution $u \in V_g$.

Mechanics reading. (VE) is "no rigid-body mode": the rim is held, so a membrane cannot move without stretching. (4b.5) is the 2D version of Lecture 8's "pinned ends cannot move for free" — held on the left side, a deflection is controlled by its slope. $\beta^*/\beta_* = 8$ is the constant that Céa's lemma will carry into the error estimate of part (d).

Problem 4(c)

Problem. Develop a finite element method for this problem based on square elements. On each element the solution on the four corners is interpolated with $s(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 xy$. Write out the basis functions and explicitly write out the linear system (uniform grid, same spacing in x and y).

Step 1 — grid, element, local coordinates

- $N \times N$ squares, $h = 2/N$, nodes $x_i = -1 + ih$, $y_j = -1 + jh$, $i, j = 0, \ldots, N$. Interior nodes $1 \le i, j \le N-1$: $(N-1)^2$ unknowns. Boundary nodes carry the known value $g(x_i, y_j)$.
- Element $K_{ij} = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$, local coordinates $\xi = (x - x_i)/h$, $\eta = (y - y_j)/h \in [0,1]$, corners numbered counter-clockwise from (x_i, y_j) (Figure 4).

Step 2 — the four corner values fix s; the shape functions

- Put $x = x_i + h\xi$, $y = y_j + h\eta$ into s and collect powers:

$$s = \underbrace{\alpha_1 + \alpha_2 x_i + \alpha_3 y_j + \alpha_4 x_i y_j}_{c_1} + \underbrace{h(\alpha_2 + \alpha_4 y_j)}_{c_2}\,\xi + \underbrace{h(\alpha_3 + \alpha_4 x_i)}_{c_3}\,\eta + \underbrace{h^2 \alpha_4}_{c_4}\,\xi\eta. \tag{4c.1}$$

  Read bottom-up, $\alpha_4, \alpha_3, \alpha_2, \alpha_1$ follow from c_4, c_3, c_2, c_1 one at a time, so the α's and the c's carry the same information.

- Evaluate (4c.1) at the four corners, $s_1 = s(0,0)$, $s_2 = s(1,0)$, $s_3 = s(1,1)$, $s_4 = s(0,1)$: $s_1 = c_1$, $s_2 = c_1 + c_2$, $s_4 = c_1 + c_3$, $s_3 = c_1 + c_2 + c_3 + c_4$. Solved one at a time,

$$c_1 = s_1, \quad c_2 = s_2 - s_1, \quad c_3 = s_4 - s_1, \quad c_4 = s_3 - s_2 - s_4 + s_1, \tag{4c.2}$$

  a unique interpolant for any four corner values.

- Put (4c.2) back into (4c.1) and collect by corner value: $s = \sum_{a=1}^{4} s_a N_a(\xi, \eta)$ with

$$\boxed{N_1 = (1-\xi)(1-\eta), \qquad N_2 = \xi(1-\eta), \qquad N_3 = \xi\eta, \qquad N_4 = (1-\xi)\eta} \tag{4c.3}$$

  Each is 1 at its own corner and 0 at the other three.

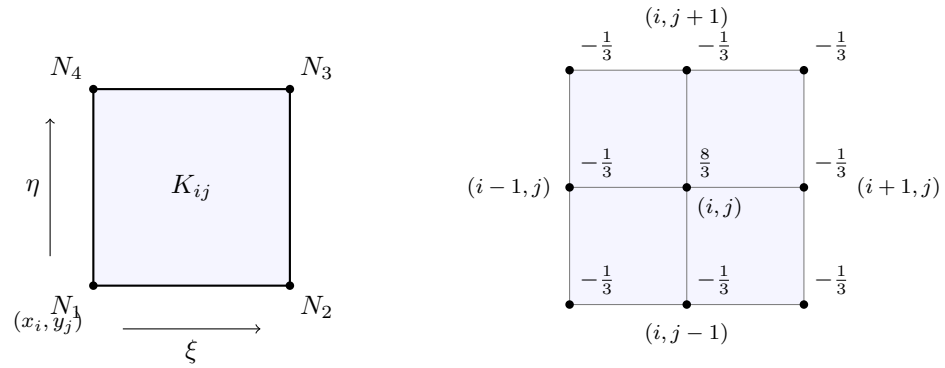


Figure 4: Left: one element and its four shape functions (4c.3). Right: the four elements around an interior node (i,j) — the support of φ_{ij} — with the coefficients of its row of the stiffness matrix, (4c.10).

Step 3 — continuity, and the global basis [Lecture 4: 1D hats; HW3 Problem 3: a jump puts a delta in the derivative]

- On the right edge of K_{ij}, $\xi = 1$ and (4c.3) gives $s = s_2(1-\eta) + s_3\eta$: only the two corner values on that edge enter. The neighbour $K_{i+1,j}$ produces the same line from the same two values on its left edge. So s is continuous across every edge, ∇s has no delta, $\iint_\Omega |\nabla s|^2 < \infty$: the method is **conforming**, $V_h \subset$ the spaces of (4a.6).
- Glue the N_a of the four squares around node (i,j) (Figure 4, right). On K_{ij} the node is corner 1, value $(1-\xi)(1-\eta)$; on $K_{i-1,j}$ it is corner 2, value $\xi(1-\eta)$; and so on. In every case the x-factor is Lecture 4's hat φ_i on the x-grid and the y-factor is φ_j on the y-grid:

$$\boxed{\varphi_{ij}(x,y) = \varphi_i(x)\,\varphi_j(y), \qquad \varphi_i(x) = \begin{cases} 1 - |x - x_i|/h, & |x - x_i| \le h, \\ 0, & \text{otherwise.} \end{cases}} \tag{4c.4}$$

- Trial and test functions, with the boundary values as Lecture 5's lift:

$$U = \underbrace{\sum_{(k,l)\in\partial\Omega} g(x_k, y_l)\,\varphi_{kl}}_{\text{known, carries } u=g} + \sum_{i,j=1}^{N-1} \xi_{ij}\,\varphi_{ij}, \qquad v = \varphi_{ij}, \quad 1 \le i,j \le N-1. \tag{4c.5}$$

(FEM) Find U of the form (4c.5) such that $\displaystyle\iint_\Omega \nabla U \cdot \nabla\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y = \iint_\Omega f\varphi_{ij}\,\mathrm{d}x\,\mathrm{d}y$ for every interior node (i,j).

Step 4 — the element stiffness matrix

- $\partial/\partial x = (1/h)\,\partial/\partial\xi$, $\partial/\partial y = (1/h)\,\partial/\partial\eta$ and $\mathrm{d}x\,\mathrm{d}y = h^2\,\mathrm{d}\xi\,\mathrm{d}\eta$, so

$$k^e_{ab} = \iint_K \nabla N_a \cdot \nabla N_b\,\mathrm{d}x\,\mathrm{d}y = \underbrace{\tfrac{1}{h^2}\cdot h^2}_{=1} \int_0^1\!\!\int_0^1 \left(N_{a,\xi}N_{b,\xi} + N_{a,\eta}N_{b,\eta}\right)\mathrm{d}\xi\,\mathrm{d}\eta. \tag{4c.6}$$

  h cancels: in 2D the stiffness of a square does not depend on its size.
- Every N_a in (4c.3) is a product of $1-\xi$ or ξ with $1-\eta$ or η, so (4c.6) splits into 1D integrals:

$$\int_0^1 (1-\xi)^2\mathrm{d}\xi = \tfrac{1}{3}, \quad \int_0^1 \xi(1-\xi)\,\mathrm{d}\xi = \tfrac{1}{2} - \tfrac{1}{3} = \tfrac{1}{6}, \quad \left[(1-\xi)'\right]^2 = 1, \quad (1-\xi)'\,\xi' = -1. \tag{4c.7}$$

- Three kinds of pair, each "x-part $\times$ y-part + y-part $\times$ x-part":

$$\underbrace{1\cdot\tfrac{1}{3} + \tfrac{1}{3}\cdot 1 = \tfrac{4}{6}}_{a=b}, \qquad \underbrace{(-1)\cdot\tfrac{1}{3} + \tfrac{1}{6}\cdot 1 = -\tfrac{1}{6}}_{a,b \text{ share an edge}}, \qquad \underbrace{(-1)\cdot\tfrac{1}{6} + \tfrac{1}{6}\cdot(-1) = -\tfrac{2}{6}}_{a,b \text{ opposite}}, \tag{4c.8}$$

$$\boxed{\boldsymbol{k}^e = \frac{1}{6}\begin{bmatrix} 4 & -1 & -2 & -1 \\ -1 & 4 & -1 & -2 \\ -2 & -1 & 4 & -1 \\ -1 & -2 & -1 & 4 \end{bmatrix}} \tag{4c.9}$$

  Each row sums to 0: a constant has no gradient, hence no energy.

Step 5 — the linear system [Lecture 13, eq. (1.236): $A_{ij} = B(\varphi_i, \varphi_j)$; Lecture 5, eq. (1.66): known values to the RHS; Lecture 4: trapezoidal load]

- **Row of an interior node** (i, j)**.** It sits in 4 squares as a corner, shares each edge-neighbour with 2 squares and each diagonal neighbour with 1. From (4c.8): $4 \cdot \frac{4}{6} = \frac{8}{3}$, $2 \cdot (-\frac{1}{6}) = -\frac{1}{3}$, $1 \cdot (-\frac{2}{6}) = -\frac{1}{3}$:

$$\boxed{\frac{1}{3}\Big[\,8\,U_{i,j} - \sum_{(k,l)\in\mathcal{N}_{ij}} U_{k,l}\Big] = b_{ij}, \qquad 1 \le i, j \le N-1,} \tag{4c.10}$$

  $\mathcal{N}_{ij}$ the eight neighbours $k \in \{i-1, i, i+1\}$, $l \in \{j-1, j, j+1\}$, $(k,l) \neq (i,j)$, and $U_{k,l} = g(x_k, y_l)$ whenever (k,l) is a boundary node.

- **Load.** $b_{ij} = \iint_\Omega f\varphi_{ij}$. The trapezoidal rule of Lecture 4 on each square, $\iint_K F \approx \frac{h^2}{4}\sum_{\text{corners}} F$, sees $f\varphi_{ij}$ only at corner (i, j), where $\varphi_{ij} = 1$; four squares:

$$b_{ij} \approx 4 \cdot \tfrac{h^2}{4}\, f(x_i, y_j) = h^2 f_{ij}. \tag{4c.11}$$

- **Matrix form.** Order the unknowns row by row, $\boldsymbol{U} = (U_{1,1}, \dots, U_{N-1,1}, U_{1,2}, \dots, U_{N-1,N-1})^T$. Move every boundary value in (4c.10) to the right:

$$\boxed{\begin{gathered}\boldsymbol{A}\,\boldsymbol{U} = \boldsymbol{b} + \boldsymbol{r}, \qquad \boldsymbol{A} = \frac{1}{3}\begin{bmatrix} \boldsymbol{T} & \boldsymbol{S} & & \\ \boldsymbol{S} & \boldsymbol{T} & \ddots & \\ & \ddots & \ddots & \boldsymbol{S} \\ & & \boldsymbol{S} & \boldsymbol{T} \end{bmatrix}, \\ \boldsymbol{T} = \begin{bmatrix} 8 & -1 & & \\ -1 & 8 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 8 \end{bmatrix}, \qquad \boldsymbol{S} = \begin{bmatrix} -1 & -1 & & \\ -1 & -1 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & -1 \end{bmatrix}\end{gathered}} \tag{4c.12}$$

  $\boldsymbol{A}$ is $(N-1)^2 \times (N-1)^2$, made of $N-1$ block rows; $\boldsymbol{T}, \boldsymbol{S}$ are $(N-1) \times (N-1)$; $\boldsymbol{T}$ couples a node to its own grid row, $\boldsymbol{S}$ to the rows above and below.

- $r_{ij} = \frac{1}{3}\sum g(x_k, y_l)$ over the boundary nodes among the eight neighbours of (i, j). For $g = 1$ (part (d)):

$$r_{ij} = \begin{cases} 5/3, & (i,j) \text{ next to a corner (5 boundary neighbours)}, \\ 1, & (i,j) \text{ next to one side (3 boundary neighbours)}, \\ 0, & \text{otherwise.} \end{cases} \tag{4c.13}$$

- $\boldsymbol{A}$ is symmetric positive definite by Lecture 13's claim: (S) and (VE) of 4(b) hold, whatever the element shape.

Same as finite differences? Divide (4c.10) by h^2 and use (4c.11): $\big[8U_{i,j} - \sum_{\mathcal{N}_{ij}} U_{k,l}\big]/(3h^2) = f_{ij}$, a nine-point difference formula for $-\Delta u$. This is Lecture 4's "CG(1) with trapezoidal load is the central difference method", one dimension up: the 2D element couples diagonal neighbours too, which the five-point formula does not.

Problem 4(d)

Problem. Use the method of 4(c) to solve $-\nabla\cdot\nabla u = f$ on $\Omega=[-1,1]^2$, $u=g$ on $\partial\Omega$, with

$$f=\pi^2 e^{\sin(\pi x)\sin(\pi y)}\big\{2\sin(\pi x)\sin(\pi y)+2\cos^2(\pi x)\cos^2(\pi y)-\cos^2(\pi x)-\cos^2(\pi y)\big\},\qquad g=1.$$

Create a convergence table showing both $\|\cdot\|_V$ and $\|\cdot\|_{L^2(\Omega)}$ errors. What order of convergence do you observe? The exact solution is $u=e^{\sin(\pi x)\sin(\pi y)}$.

Step 1 — the data are consistent

- With $s=\sin\pi x\sin\pi y$ and $u=e^s$, $u_{,x}=\pi\cos\pi x\sin\pi y\,e^s$ and

$$\begin{aligned}-u_{,xx}-u_{,yy}&=\pi^2e^s\big[2\sin\pi x\sin\pi y-\cos^2\pi x\ \underbrace{\sin^2\pi y}_{1-\cos^2\pi y}-\underbrace{\sin^2\pi x}_{1-\cos^2\pi x}\cos^2\pi y\big]\\&=\pi^2e^s\big[2\sin\pi x\sin\pi y+2\cos^2\pi x\cos^2\pi y-\cos^2\pi x-\cos^2\pi y\big]=f.\end{aligned}\tag{4d.1}$$

  Also checked symbolically ($-\Delta u-f\equiv 0$).
- On every side $\sin(\pm\pi)=0$, so $s=0$ and $u=e^0=1=g$.

Step 2 — the implementation [4(c); Lecture 5, eq. (1.66)]

- **Grid.** $N\times N$ squares, $N=4,\dots,256$, $h=2/N$, $(N-1)^2$ unknowns.
- **Stiffness.** The element matrix $\boldsymbol{k}^e$ of 4(c), the same on every square, assembled. Checked in the code: every interior row is the stencil $\frac{1}{3}[8;\,-1\times 8]$ of 4(c), the interior matrix is symmetric positive definite, and the boundary terms are 5/3, 1, 0 as in 4(c).
- **Boundary.** $U=g=1$ at boundary nodes, moved to the right-hand side (Lecture 5).
- **Load.** $b_{ij}=\iint f\varphi_{ij}$ by 3×3 Gauss points per square, not the trapezoidal h^2f_{ij} of 4(c); Step 4 compares.
- **Errors,** 4×4 Gauss points per square, with the norm of Lecture 13, eq. (1.254):

$$\|e\|^2_{L^2(\Omega)}=\iint_\Omega (u-U)^2\,\mathrm{d}x\,\mathrm{d}y,\qquad \|e\|^2_V=\|e\|^2_{L^2(\Omega)}+\iint_\Omega|\nabla(u-U)|^2\,\mathrm{d}x\,\mathrm{d}y.\tag{4d.2}$$

  Rate $=\log_2\big(E(h)/E(h/2)\big)$.

Step 3 — convergence table

N	h	unknowns	$\|e\|_V$	rate	$\|e\|_{L^2(\Omega)}$	rate
4	0.50000	9	2.1378×10^{0}	—	2.3519×10^{-1}	—
8	0.25000	49	1.4636×10^{0}	0.55	9.5737×10^{-2}	1.30
16	0.12500	225	7.3914×10^{-1}	0.99	2.4256×10^{-2}	1.98
32	0.06250	961	3.7048×10^{-1}	1.00	6.0808×10^{-3}	2.00
64	0.03125	3969	1.8536×10^{-1}	1.00	1.5213×10^{-3}	2.00
128	0.01562	16129	9.2693×10^{-2}	1.00	3.8038×10^{-4}	2.00
256	0.00781	65025	4.6348×10^{-2}	1.00	9.5100×10^{-5}	2.00

Table 7: Bilinear squares on $[-1,1]^2$, $N\times N$ elements.

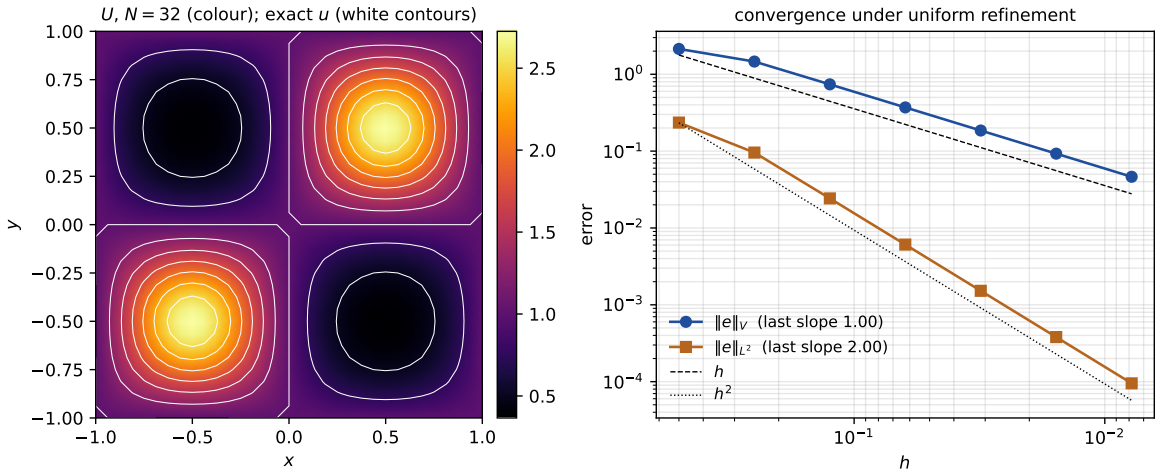


Figure 5: Left: U at $N = 32$ with contours of the exact u. Right: the errors of Table 7 against h, with slopes 1 and 2 for reference.

Step 4 — the observed orders

- $\|e\|_V$: rate $\to 1.00$, i.e. $\mathcal{O}(h)$. $\|e\|_{L^2(\Omega)}$: rate $\to 2.00$, i.e. $\mathcal{O}(h^2)$. Clean from $N = 16$; at $N = 4, 8$ the mesh has only two to four squares across each bump of u.
- The trapezoidal load of 4(c) gives the same rates; at $N = 256$, $\|e\|_V = 4.6355 \times 10^{-2}$ and $\|e\|_{L^2} = 9.7057 \times 10^{-5}$, against 4.6348×10^{-2} and 9.5100×10^{-5} with Gauss.
- **Why $\mathcal{O}(h)$ in $\|\cdot\|_V$.** $g = 1$ is reproduced exactly by the boundary terms of 4(c), so $U - 1$ is the Galerkin approximation of $w = u - 1 \in H^1_0$, and Céa's lemma (Lecture 13, eq. (1.242)) with the constants of 4(b) gives

$$\|u - U\|_V \le \frac{\beta^*}{\beta_*} \|u - v\|_V = 8 \|u - v\|_V \qquad \text{for every admissible } v \in V_h. \tag{4d.3}$$

  With v the bilinear interpolant of u, the right side is $\mathcal{O}(h)$.
- **Why $\mathcal{O}(h^2)$ in L^2.** Lecture 7's duality argument: the interpolation estimate is used twice, once on e and once on the dual solution, one factor of h each.

NOT FROM LECTURE — the interpolation estimate for squares

Lecture 13 states $\|u - u_h\|_{H^1} \le c \|h\, u\|_{H^2}$, eq. (1.262), for linear triangles. The same $\mathcal{O}(h)$ bound for the bilinear interpolant on squares, and the 2D duality argument behind $\mathcal{O}(h^2)$ in L^2, are standard but were not shown in class. Table 7 confirms both rates.

Conclusion. Bilinear squares converge at $\mathcal{O}(h)$ in $\|\cdot\|_V$ and $\mathcal{O}(h^2)$ in $\|\cdot\|_{L^2(\Omega)}$ — the same orders as CG(1) — with $\|e\|_V = 4.63 \times 10^{-2}$ and $\|e\|_{L^2} = 9.51 \times 10^{-5}$ at $N = 256$.